ALAB 351.2 — Data Types, Variables, Operators, and Basic I/O

James Sloan · UCI 3052 Python Essentials · Module 351 · August 18, 2026

Repository

Repository: github.com/santigrey/Python-Essentials

Lab folder: modules/Module 351/ALAB-351.2

All three scripts and a README containing example output are in that folder.

File	Task
`types_and_vars.py`	Part 1 — variables, data types, operators
`simple_calculator.py`	Part 2 — two-number calculator
`string_fun.py`	Bonus — string operations
`README.md`	Example output from running each script

Example Output

`types_and_vars.py`

Takes no input, so the output is the same every run.

```
Section 1: Variables and their data types
-----------------------------------------
name   = 'James'  -> str
age    = 41      -> int
height = 1.83    -> float

Section 2: Introduction
-----------------------
Hello, my name is James. I am 41 years old and 1.83 meters tall.

Section 3: Age in five years
----------------------------
In 5 years, I will be 46 years old.
```

```
Section 4: Rectangle area
-------------------------
The area of a 5.5 x 2 rectangle is 11.0.

Section 5: Operators
--------------------
age + 5     = 46
age - 5     = 36
age * 2     = 82
age / 2     = 20.5
age // 2    = 20
age % 2     = 1
height ** 2 = 3.3489
"James" + " " + "Sloan" = "James Sloan"
"-" * 20 = "--------------------"
"James Sloan".upper() = "JAMES SLOAN"
len("James Sloan") = 11
```

`simple_calculator.py`

Full transcript of one run:

```
Simple Calculator
=================
Enter the first number: 7
Enter the second number: 3
Choose an operation (+, -, *, /): *

7 * 3 = 21
```

The other three operations, run the same way:

First	Second	Operation	Output
`15.5`	`4`	`-`	`15.5 - 4 = 11.5`
`-12`	`4.25`	`+`	`-12 + 4.25 = -7.75`
`10`	`3`	`/`	`10 / 3 = 3.3333`

Invalid input is re-prompted instead of crashing, and division by zero is caught:

```
Enter the first number: abc
  'abc' is not a number. Please try again.
Enter the first number: 8
Enter the second number: 0
Choose an operation (+, −, *, /): /

8 / 0 = undefined
Error: division by zero is not allowed.
```

`string_fun.py`

```
Enter a word: Python

Length:        6 characters
Uppercase:     PYTHON
Repeated x3:   Python Python Python
Lowercase:     python
Title case:    Python
First letter:  P
Last letter:   n
Reversed:      nohtyP
Palindrome:    no
```